

Introduction

- **Greenhouse gases** absorb heat and **trap** it in the atmosphere (A. K et al., 2023).
- Increased atmospheric temperatures lead to increased **surface water** temperatures.
- Climate change and human activity is leading to **decreased biodiversity** within ecosystems.
- Since *Lymnaea stagnalis* are often found in ponds, increased water temperatures could impact their **health** and **development**, making this research **vital for future generations**.



Lymnaea stagnalis egg mass (image taken by researcher)

- The **CDCH-1** gene plays a key role in **egg-laying behavior** in *Lymnaea stagnalis*.
- Studies have found that the Caudodorsal Cell Hormone causes **egg mass formation** and triggers the last two phases of egg laying behavior (Hermann et al., 1997).

Purpose

The purpose of this experiment was to determine the effects of increased water temperatures on the expression of CDCH-1 in *Lymnaea stagnalis* (the common pond snail). Global warming has been relatively understudied in terms of embryonic development, especially in *Lymnaea stagnalis*.

It was hypothesized that if *L. stagnalis* are exposed to increasing temperatures, then those exposed to slightly higher temperatures will have increased reproduction rates due to improved metabolic activity, but those that are exposed to extremely higher temperatures will experience detrimental effects to their reproduction due to excess physiological stress.

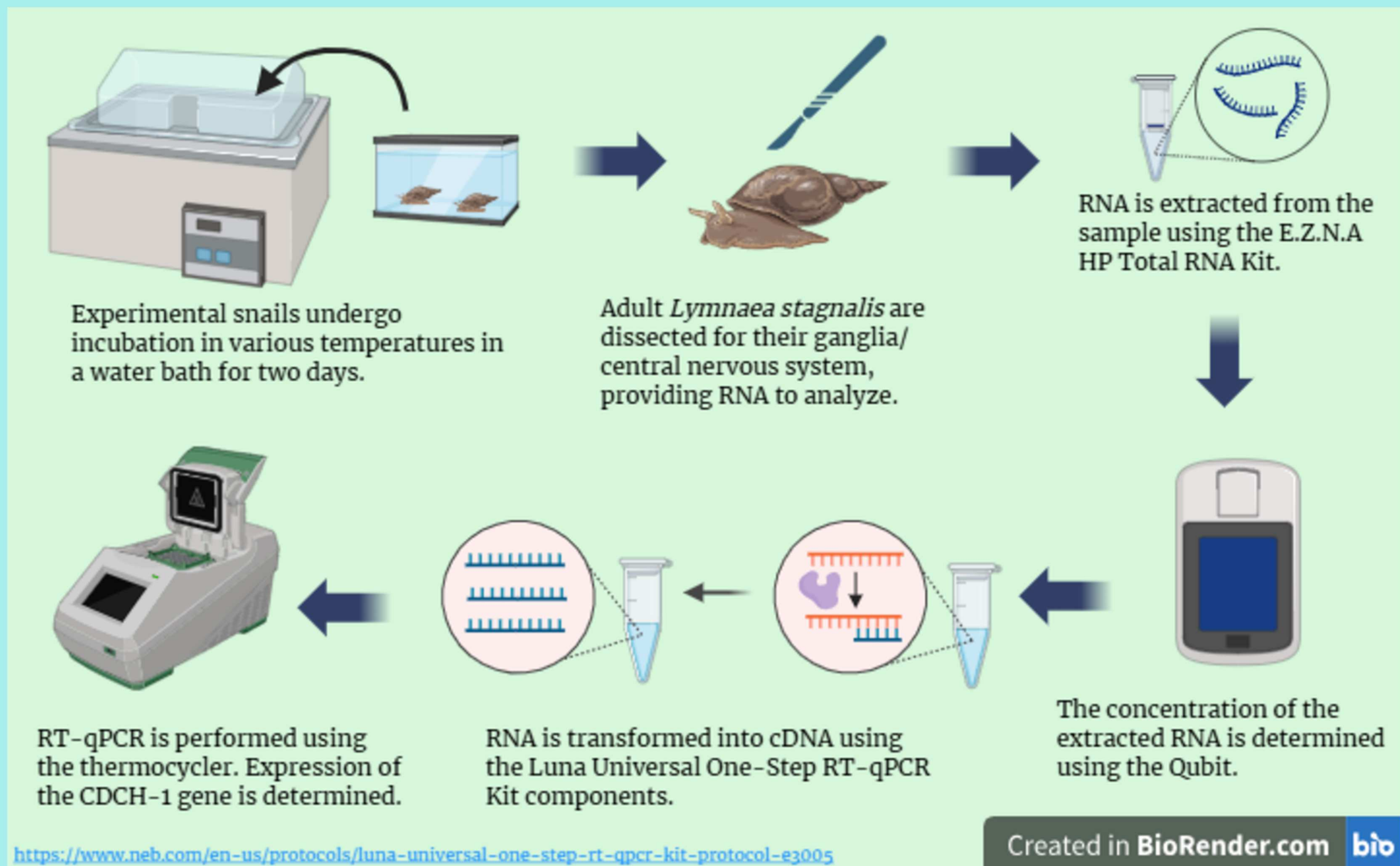
Previous Study

Wang et al. (2015)

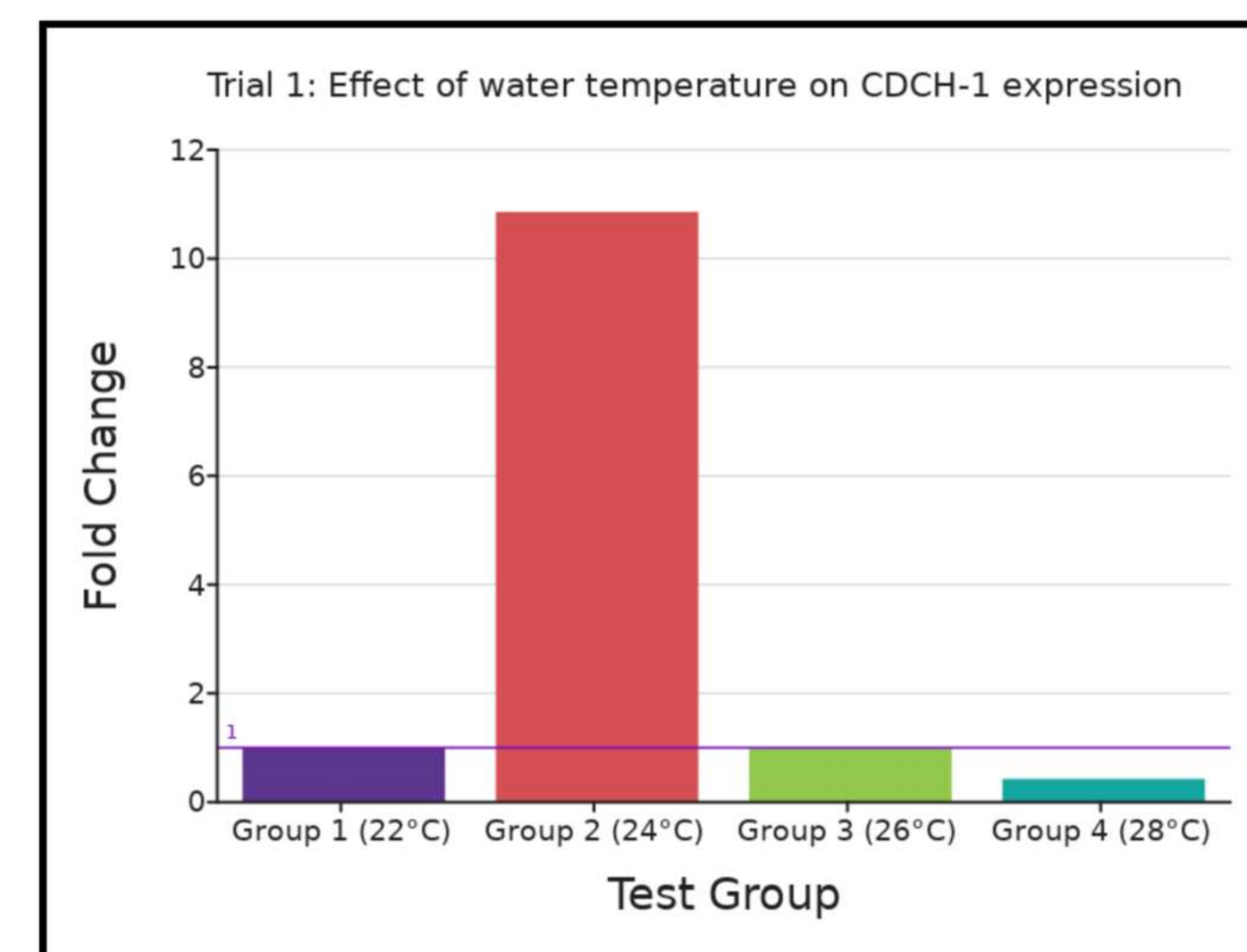
- Studied the impacts of varying temperatures and pH on the pufferfish *Takifugu obscurus* eggs.
- They observed slight increases in temperature increased hatching rate, but when temperatures exceeded the optimal range, hatching rate significantly declined.

Determining the impact of rising water temperatures on the fecundity and expression of the Caudodorsal Cell Hormone (CDCH-1) in *Lymnaea stagnalis*

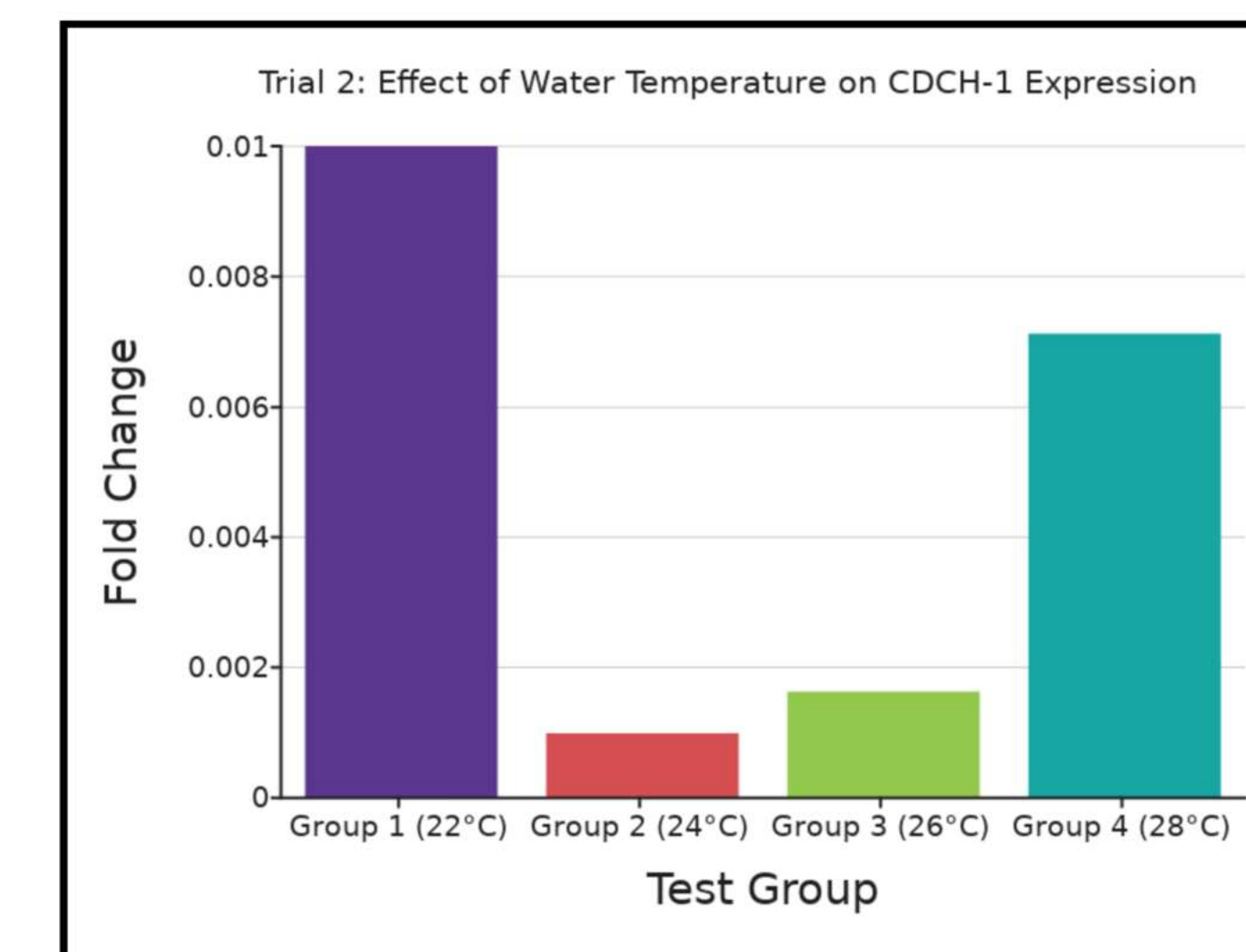
Methods



Results



Trial 1 results aligned with my hypothesis. When temperature was slightly increased, CDCH-1 was largely upregulated. Then, when temperature was extremely increased, CDCH-1 was downregulated.

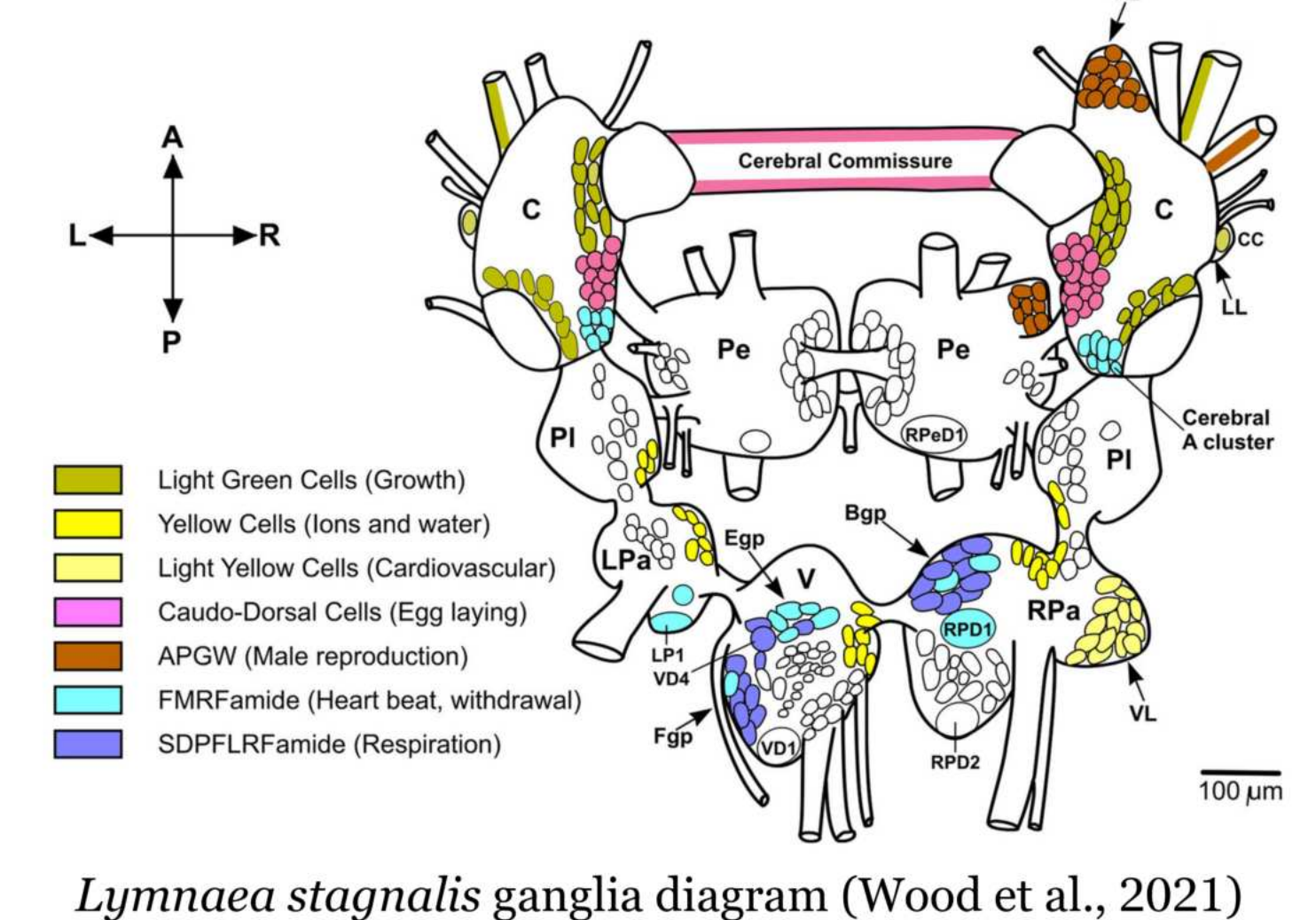


Trial 2 results did not align with my hypothesis. In trial 2, any increase in temperature resulted in a very large CDCH-1 downregulation.

*Note that the reference/control for fold change is 1, however the Y-axis is scaled for visibility purposes.

Conclusions

- Overall, **not enough data** was collected to make sound conclusions.
- Trial 1 data showed a **clear trend** that matches previous studies.
 - When temperature was increased by 2°C, CDCH-1 was **upregulated** by 1085.7%.
 - When temperature was increased by 6°C, CDCH-1 was **downregulated** by 42.1%.
- Trial 2 data did not have a clear trend and **did not align** with trial 1 data.
 - Each experimental group had a fold change percentage of <1%, indicating a **very large downregulation**.
 - These results could be explained by snails being of older age than trial 1 or errors in the RT-qPCR processes.
 - These results indicate that any increase in temperature leads to **determinantal impacts on reproductive health**.
- In order to improve results, **more trials** should be run, with a **larger sample size** and **longer test trials**.



Lymnaea stagnalis ganglia diagram (Wood et al., 2021)

Future Research

In the future, more research could be done regarding generational impacts of increased water temperatures. Studies could be done on the effect of high water temperatures on snail reproduction over their whole life.

Additionally, more research should be done regarding other aquatic organisms to gain a more holistic view of how rising water temperatures are impacting the whole ecosystem.

References

- A. K. P., M. M., Rajamanickam, S., Sivarethinamohan, S., Gaddam, M. K. R., Velasamy, P., R. G., Ravindran, G., Gurugubelli, T. R., & Muniasamy, S. K. (2023). Impact of climate change and anthropogenic activities on aquatic ecosystem - A review. *Environmental Research*, 238. <https://doi.org/10.1016/j.envres.2023.117233>
- Hermann, P. M., de Lange, R. P., Pieneman, A. W., ter Maat, A., & Jansen, R. F. (1997). Role of neuropeptides encoded on CDCH-1 gene in the organization of egg-laying behavior in the pond snail, *Lymnaea stagnalis*. *Journal of Neurophysiology*, 78(6), 2859-2869.
- Koene, J. M., Jackson, D. J., Nakadera, Y., Cerveau, N., Madoui, M.-A., Noel, B., Jamilloux, V., Poulain, J., Labadie, K., Da Silva, C., Davison, A., Feng, Z.-P., Adema, C. M., Klopp, C., Aury, J. M., Winkler, P., & Coste, M.-A. (2024). The genome of the simultaneously hermaphroditic snail *Lymnaea stagnalis* reveals an evolutionary expansion of FMRFamide-like receptors. *Scientific Reports*, 14(1). <https://doi.org/10.1038/s41598-024-78520-1>
- Wang, J., Li, Z., Chen, Y., & Yang, Z. (2015). The combined effect of temperature and pH on embryonic development of obscure puffer *Takifugu obscurus* and its ecological implications. *Biochemical Systematics and Ecology*, 58, 1-6. <https://doi.org/10.1016/j.bse.2014.10.012>
- Wood, E. A., Stophka, S. A., Zhang, L., Mattson, S., Mass, G., Pirger, Z., & Vortes, A. (2021, May 19). Neuropeptide localization in *Lymnaea stagnalis*: From the central nervous system to subcellular compartments. *Frontiers*. <https://www.frontiersin.org/journals/molecular-neuroscience/articles/10.3389/fnmol.2021.670303/full>